

# TCS3351 CRYPTOGRAPHY AND DATA SECURITY

## Tutorial 8 - Modern Asymmetric Key Ciphers: Principles

1. State the major differences between symmetric-key and asymmetric-key cryptosystems.

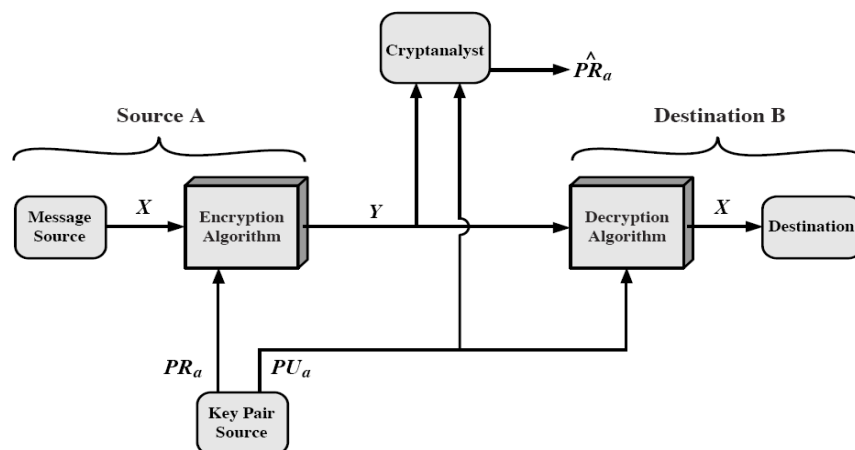
Answer:

**Symmetric-key cryptosystem:** Uses only one private key in both encryption and decryption processes. It does not ensure integrity and non-repudiation properties. That is, it does not protect the sender from the receiver forging the message (or signature) or claiming the message (or signature) sent by the sender.

**Asymmetric-key cryptosystem:** Uses a pair of public and private keys, with the public key used in encryption and verification. The private key used in decryption and signing processes. It ensures integrity and non-repudiation in communication.

2. With the help of a diagram, explain how the public-key cryptosystem can be used to provide authentication/digital signature.

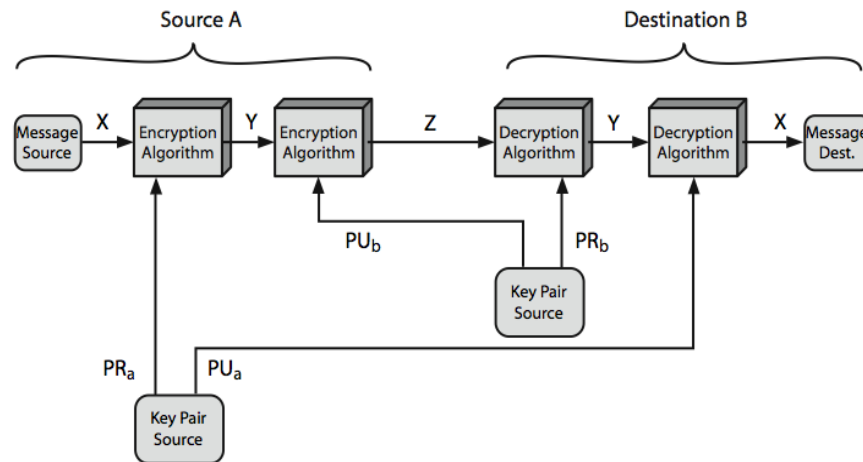
Answer:



In a public-key cryptosystem, a message signer  $A$  will sign the message using his private key  $\sigma = \text{Sig}(\text{Prv}_A, M) = \text{Enc}(\text{Prv}_A, M)$ , which the signer himself can only prepare. When the signature is sent to the receiver, he can verify the authenticity and validity of the signature using the signer's public key  $M = \text{Vfy}(\text{Pub}_A, \sigma) = \text{Dec}(\text{Pub}_A, \sigma)$ , which can be done by anyone who receives the signature.

3. With the help of a diagram, explain how the public-key cryptosystem can be used to provide both confidentiality and authentication.

Answer:



Both confidentiality and authentication can be achieved using a hybrid public-key cryptosystem, in which two pairs of public-private keys are used. A pair of public-private keys from the sender A, ( $Prv_A$ ,  $Pub_A$ ) will be used in signing-verifying the message, while a pair of public-private keys from the receiver B, ( $Prv_B$ ,  $Pub_B$ ) will be used in encrypting-decrypting the message.

The message is firstly signed using  $Prv_A$  and next encrypted using  $Pub_B$ . When the message is received, it will be decrypted using  $Prv_B$  and finally verify using  $Pub_A$ .  
Real case: B4 signing, you need to hash the msg then padded it.

>>>

- Alice 1 2 send a signed small document to Bob and only Bob can see the document.
- Sign 1<sup>st</sup>. Encrypt Later.
- Alice key is 2048 bits
  - $\text{Sign}(\text{doc}, \text{AlicePriv}) = \text{signature}$ . 256 bytes
  - $\text{Encrypt}(\text{RSA}, \text{signature}, \text{BobPub}) = \text{CipherTxt}$ . 512 bytes \*\*\*
  - Send CipherTxt to Bob.
- Bob key is 4096 bits.
  - $\text{Decrypt}(\text{RSA}, \text{Ciphertxt}, \text{BobPriv}) = \text{Signature}$
  - $\text{Verify}(\text{Signature}, \text{AlicePub}) = \text{True/False}$ . If True, he will get the doc.

4. Briefly explain the One-way and Trap-door one-way functions.

Answer:

One-way function: A function maps every object in a domain into its corresponding unique image in the range, such that the function is easily computable but infeasible (or difficult) to invert it. Known  $x$ , find  $y = f(x)$  is easy. Known  $y$ , find  $(x)$  is extremely hard.

Trap-door one-way function: A function maps every object in a domain into its corresponding unique image in the range, such that the function is easily computable but infeasible (or difficult) to invert it, **except when trap-door information is given.**

**References:**

1. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.
2. William Stallings, “Cryptography and Network Security – Principles and Practice”, Seventh Edition, Prentice Hall, 2017